

**KAUA'I BEEF SUPPLY AND CONVERSION CAPACITY: TESTING MIDSTREAM
CONSTRAINTS AGAINST UPSTREAM EXPORT REALITIES**

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KAUA'I BEEF SUPPLY AND CONVERSION CAPACITY: TESTING MIDSTREAM CONSTRAINTS AGAINST UPSTREAM EXPORT REALITIES

1. INTRODUCTION AND PROBLEM FRAMING

Kaua'i maintains 42,345 acres of pasture (HDOA, 2020) and 13,893 cattle and calves (USDA, 2022), but Hawai'i imports 85% to 90% of its food (Leung & Loke, 2008). Policy discourse frequently attributes this local beef deficit to downstream conversion bottlenecks—scarcity of slaughterhouse infrastructure, labor, and cold storage. Under this framing, maritime shipping disruptions are treated as exogenous shocks that cause shortages, rather than stress events that expose existing domestic conversion failures. This study evaluates whether downstream capacity truly acts as the primary constraint by benchmarking throughput against upstream herd allocations and market flows.

2. METHODOLOGY & FIELD EVIDENCE

Herd distributions and labor rates were established using the USDA 2022 Census of Agriculture and BLS OEWS May 2025 nonmetropolitan data. Primary midstream operational constraints were identified via audits of Kaua'i's main multi-user inspected processing facility (Operations manager, personal communications, October 1 and 2, 2026). According to the operations manager, other island facilities serve closed channels or remain inactive for regular commercial slaughter, positioning the audited plant as the primary gateway for independent producers.

3. EMPIRICAL FINDINGS: TESTING THE CAPACITY HYPOTHESES

3.1 Midstream Processing: Shallow Slack and Fixed Infrastructure Ceilings

Field evidence indicates the facility exhibits operational slack, but it is shallow and constrained by fixed infrastructure:

- Queue Friction & Utilization: Independent producers face ~1-week lead times.

Management reports slaughtering ~8 head on a Wednesday shift and a similarly sized weekend shift (~16 head per week). Operating frequency is governed strictly by inbound deliveries.

- **The Shallow Capacity Ceiling:** The facility's maximum throughput under existing infrastructure is ~25 head per week, leaving only ~9 head of unutilized slack. As illustrated in Figure 1, this represents a severe midstream bottleneck when compared to the ~71 calves leaving the island weekly. Expanding beyond this ceiling is restricted by physical capital. Higher volume substantially increases wastewater volume and tank pumping, electricity for continuous refrigeration, and remote maintenance friction for equipment wear. Additionally, physical hanging rail space creates a hard capacity limit during peak weekend harvests.
- **Labor Structure:** Butchering runs part-time (3:00–6:00 AM) to accommodate staff's primary employment. Expanding to full-time requires competitive wages (BLS reports an \$18.90 median hourly wage for Hawai'i/Kaua'i nonmetro slaughterers, lagging the \$20.65+ resort baseline), which low fixed processing revenues cannot currently support.
- **Regulatory Friction:** USDA FSIS inspection is reliable and does not restrict harvest schedules.

3.2 The Upstream Structural Divergence: Comparative Advantage and Opportunity Cost

Because the facility operates with unutilized capacity, the primary supply constraint is upstream. Kaua'i marketed 3,707 calves under 500 lbs and 1,911 cattle 500+ lbs (USDA, 2022). Calves represent ~66% of marketing, an annualized outflow of up to 71.3 calves per week (an upper bound including local transfers). The driver is comparative advantage: continental commercial feedlots leverage cheap, energy-dense grain rations, while Kaua'i's endowment favors extensive cow-calf pasture grazing. This creates a steep opportunity cost:

- **Route A (Export):** October 2026 continental auction prices for 400-500 lb steers hit \$4.70-\$5.47/lb (USDA AMS, 2026). At a conservative \$5.00/lb, a 450 lb calf grosses \$2,250. Deducting an assumed ~\$200 for freight, the immediate net cash settlement is ~\$2,050 at Year 0.
- **Route B (Local Finishing):** Retaining a calf to 1,000 lbs yields a 560 lb carcass. At \$4.75/lb wholesale (Operations manager, personal communication, October 1, 2026),

gross revenue is \$2,660. Deducting \$350 for processing (Operations manager, personal communication, October 1, 2026) and an assumed \$1,200-\$1,400 for 24-month pasture carrying costs yields a nominal margin of \$910-\$1,110.

- Time and Risk Discounting: Discounting this nominal return over two years at 8% and factoring a 3% mortality risk ($1 / 1.08^2 \times 0.97 \approx 0.832$), the risk-adjusted Net Present Value (NPV) of local finishing is \$757 to \$923 per head.

The present-value deficit of local finishing is \$1,127 to \$1,293. Bridging this nominal gap at harvest requires an additional \$1,354-\$1,554 per head, or \$2.41-\$2.77/lb on a 560 lb carcass.

4. POLICY IMPLICATIONS AND THE RESILIENCE EXTERNALITY

4.1 The Economic Objection to Production Subsidies

From a free-market efficiency perspective, subsidizing local pasture finishing is allocatively inefficient. Bridging the gap to retain one-third of exported calves (1,236 head) requires an annual public subsidy of ~\$1.79 million, or ~\$4.14 per retail pound. Critics argue it would be cheaper to purchase imported beef and maintain a frozen strategic county stockpile.

4.2 The Resilience Externality Defense

The \$1.79 million subsidy is a market distortion justified by the risk diversification it buys. First, maintaining a 30-day county frozen reserve requires holding roughly 485,000 lbs of inventory. This demands steep capital warehousing outlays, high continuous electricity overhead, and inventory rotation spoilage. While a local processing plant also requires electricity, a "walking reserve" of live cattle holds its value during a prolonged hurricane grid failure, whereas a frozen stockpile spoils immediately. Phase 2 capital grants can equip the local facility with backup generation to ensure processing continues post-disaster.

Second, domestic agriculture mitigates single-sector economic vulnerability. During Kaua'i's April 2020 pandemic lockdown, unemployment surged to 34.4% (Hawai'i DLIR, 2020). Subsidizing this walking reserve provides vital non-tourism economic activity. This requires Phase 1 forward contracts to target institutional buyers (e.g., schools) whose budgets survive tourism crashes.

4.3 Sequential Policy Architecture and Coordination Problem

Policy must address a coordination problem: ranchers will not retain calves without slaughter access, while processors cannot justify lumpy capital investments without guaranteed volume. Interventions must deploy sequentially:

- Phase 1: Dynamic Contracts & Working Capital: The State must fund a dynamic price-bridge grant covering the \$2.41-\$2.77/lb premium, allowing schools to buy at wholesale (\$4.75/lb) while ranchers gross \$7.16-\$7.52/lb. Pegging this to a trailing 30-day AMS feeder price ensures the subsidy contracts if national cattle markets cool. In parallel, low-interest revolving credit lines must finance the 24-month grass-finishing working capital cycle.
- Phase 2: Immediate Infrastructure Grants: With a 25-head ceiling and 16-head current volume, slack is only ~9 head/week. Retaining just 13% of exported calves (~9 head) exhausts midstream capacity. Therefore, capital grants for wastewater, cooler rails, and backup generation must deploy simultaneously with Phase 1.

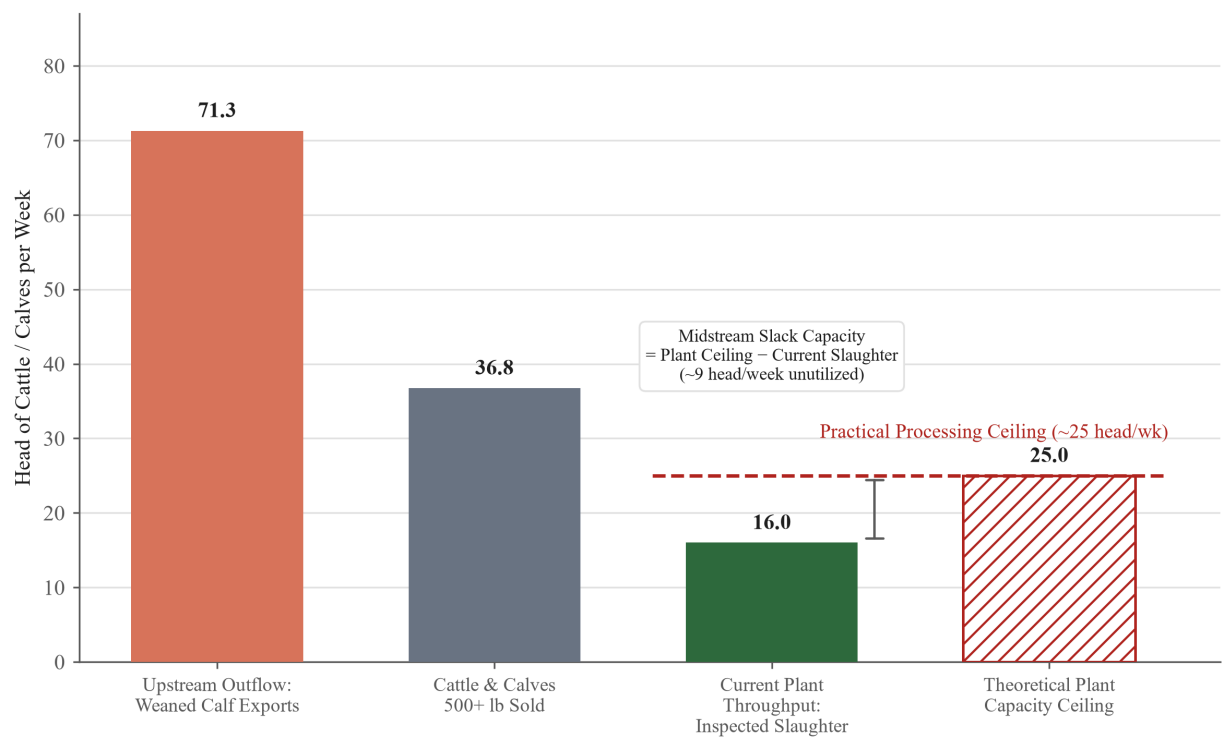
5. CONCLUSION

Kaua'i's beef deficit begins upstream due to the opportunity cost of calf exports. However, midstream infrastructure is bounded by a shallow 25-head weekly ceiling. Food security requires state-funded dynamic forward contracts to close the upstream margin gap, paired with immediate infrastructure grants to expand midstream capacity before retention volume exhausts the facility's limits.

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Figure 1: Kaua'i Weekly Bovine Allocation vs. Inspected Processing Capacity



Sources: USDA NASS 2022 Census of Agriculture, Table 11 (annual sales ÷ 52); plant values from primary operational audit (personal communications, Oct. 1–2, 2026).